

LEGAL CONNECT.

Project Proposal



**A Smart Legal Consultation & Lawyer Practice Management
Platform for Sri Lanka**

Course: SCS2202/IS2102 - Group Project I - Group No: 23

Project Proposal

for

LegalConnect

A Smart Legal Consultation and Lawyer Practice Management Platform for Sri Lanka

Course: SCS2202/IS2102 - Group Project I

Group No: 23

Date: June 2026

Table 1: Group Members

Index Number	Name of Student
24020206	D.P.J.B.D Silva
24021113	P.D.V.R Weerakoon
24020818	O.S Premachandra
24020664	P. M. P. Methsarani

Table 2: Project Personnel and Client Details

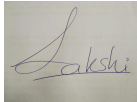
Item	Details
Proposed Project Supervisor (Academic Staff of UCSC)	M.V.P. Thilini Lakshika
Signature of the Supervisor	
Date	14/06/2026
Proposed Project Co-Supervisor (Assigned by Course Coordinator)	H. A. V. Ranujee
Signature of the Co-Supervisor	
Date	14/06/2026
Client Name	M.V.P. Thilini Lakshika (Supervisor acting as client)
Address of the Client	University of Colombo School of Computing (UCSC)
Contact Person at Client	M.V.P. Thilini Lakshika
Contact Number of the Contact Person	+94713179029
E-mail Address of the Contact Person	tlv@ucsc.cmb.ac.lk

Table of Contents

01) Introduction to Project.....	3
02) Project Goal.....	3
03) Scope of the Project.....	3
3.1 In Scope.....	4
3.2 Out of Scope.....	5
04) Objectives of the Project	5
05) Project Feasibility	5
5.1 Technical Feasibility.....	5
5.2 Schedule Feasibility.....	5
5.3 Resource Feasibility.....	6
5.4 Ethical and Legal Feasibility.....	6
06) Deliverables of the Project.....	6
07) Project Constraints and Assumptions.....	6
7.1 Constraints.....	6
7.2 Assumptions.....	7
08) Requirements.....	7
8.1 Functional Requirements.....	7
8.2 System Actors.....	11
8.3 Use Case Diagram.....	11
8.4 Use Case Description.....	11
8.5 Database Design Overview.....	15
8.6 Quality Attributes Requirements.....	16
09) Technologies to be Used.....	16
9.1 API or Library Usage with Justification.....	17
10) Project Timeline.....	17
11) Declaration.....	20

01) Introduction to Project

LegalConnect is a proposed smart legal consultation and lawyer practice management platform for the Sri Lankan legal service sector. It links clients with the right lawyers and helps lawyers manage meetings, client data, legal documents, court dates, case notes, follow-ups, document preparation, and attestation tracking through a secure digital platform. The system also features lawyer discovery, secure document sharing, legal knowledge assistance and reporting, and is designed to supplement, not replace, professional legal advice.

The project is to address the largely manual nature of legal services in Sri Lanka, where clients find it difficult to access legal assistance and lawyers waste a lot of time on administrative work. LegalConnect is intended to make legal practice more accessible, communicative, efficient and digital by reducing repetitive work and streamlining legal processes. The project is appropriate to apply system analysis, software design, database design, programming, testing and software development methodologies during the duration of the group project.

02) Project Goal

This project aims to build a secure and user friendly digital platform, which acts as a bridge between clients and lawyers in Sri Lanka and supports lawyers in managing appointments, client records, drafting documents, tracking attestations, providing legal knowledge support, offering property/land registry assistance and providing practice management tools.

The team is expected to deliver a working prototype, which includes a client web application, lawyer web/mobile assistant interface, admin portal, database system, analysis/design diagrams, documentation, and evidence of testing, by the end of the project.

03) Scope of the Project

LegalConnect will be made up of a client web app, a lawyer web app, a lawyer mobile assistant app and an admin portal. The system will enable clients to search lawyers, book appointments, upload documents, request services and receive basic legal information. Lawyers will use the system to manage profiles, appointments, clients, documents, case notes, daily diary tasks and references to legal knowledge. Admin users will verify lawyers and manage categories, manage content and monitor reports.

The system will feature innovative modules such as lawyer digital assistant, secure personalized client database, document automation with attestation tracking, legal knowledge engine, similar case recommendation and property/land registry assistance workflow. The system will support future integration with selected APIs where allowed, but the first version can work without dependency on external APIs

3.1 In Scope

- Client registration, login, and profile management.
- Lawyer Registration, Profile Management and Admin Verification based on NIC, Bar details and Supreme Court certificate details.
- Find and filter lawyers by location, court, legal category, experience, availability and rating.
- Booking, rescheduling and cancellation of appointment. Tracking of reminder and management of consultation packages.
- Client database for lawyers that includes client info, documents, case notes, appointment history, follow-up records.
- Document preparation templates, secure sharing of documents, status tracking of lawyer review and attestation.

LegalConnect will support selected structured templates for preparing preliminary legal drafts using client and case information. All generated documents will be clearly marked as drafts and must be reviewed and approved by a qualified lawyer before signing, attestation, submission, or legal use.

Supported categories include:

- Client and case administration
- Legal correspondence
- Basic agreements and commercial drafts
- Family and child-related drafts
- Affidavits and declarations
- Employment-related drafts
- Property and land-related drafts
- Civil disputes and debt recovery
- Criminal and incident records
- Other selected specialized drafts

The system will also support secure document sharing and tracking of lawyer review, signing, and attestation status.

- Legal information chatbot and legal knowledge engine using approved content for general information purposes only.
- Case law library, reference to similar cases, support for checking law overlap, legal research assistance for lawyers.
- Lawyer daily diary, meeting summary assistant, task management.
- Property and Land Registry Assistance Request Workflow with Manual Service Handling and Readiness for Future Integration.
- Admin dashboard, user management, category management, content management, reports and audit logs.
- Responsive client web interface and lawyer web/mobile assistant interface.
- Lawyer subscription payments will also be processed through the selected payment gateway.

- LegalConnect will support online payments for consultation packages, appointments, and other legal services, with payment amount, reference, date, and status recorded by the system.

3.2 Out of Scope

- Final legal advice automatically generated without review by a lawyer.
- Automated judicial decision making or prediction of court judgment.
- Direct integration with court systems or government legal databases in the first release.
- Full automation of execution of final legal documents that require physical signing, witnessing, notary attestation or court procedure.
- Not replace lawyers, notaries, court personnel, land registry personnel, or legal authorities.
- Real time voice to text transcription unless a future approved API is added.

04) Objectives of the Project

1. To offer a digital platform for clients to search and connect with appropriate lawyers in Sri Lanka.
2. Easier booking, rescheduling, cancelling lawyer appointments, and keeping track of appointment history.
3. Allow verified lawyers to update professional profiles, availability, consultation packages and practice information.
4. To have a secure custom client database for lawyers so that there is less chance of missing or untidy client records.
5. To facilitate drafting, secure sharing, lawyer review and attestation tracking without replacing legal professionals
6. To develop a lawyer assistant module for Schedule management, Daily diary, Case notes, Meeting summaries, Similar case reference and Reports.
7. To offer basic legal information through a controlled legal knowledge engine and chatbot with clear disclaimers.
8. To facilitate requests for assistance in property and land registry through manual workflow and to prepare for future official integration.
9. To produce useful reports and analytics for lawyers and administrators.

05) Project Feasibility

5.1 Technical Feasibility

The developed system is implemented using HTML, CSS, JavaScript, PHP and MySQL. This software combination is ideal for a year-long university group project and includes web design, database management, form processing, file uploading, user authentication and reporting. This aspect of providing services to lawyers by their mobile phones can be achieved by developing a mobile version of the website or a mobile application.

5.2 Schedule Feasibility

The project can be finished within an academic year if the development is done in phases. The basic modules such as authentication, profiles, appointment booking and document sharing should be done first. Once the core system is stable, we can then build more advanced modules, such as a lawyer assistant, a legal knowledge engine and a property assistance system.

5.3 Resource Feasibility

- There are 4 members on the team.
- Supervision will be provided by a supervisor and co-supervisor.
- Consult at least one practicing lawyer to validate the legal workflow.
- Personal laptops XAMPP VS code GitHub can be used.

5.4 Ethical and Legal Feasibility

- client confidentiality is to be maintained.
- Legal information chatbot content should be general information, not final legal advice.
- Public listing requires verification by a lawyer.
- Lawyers need to vet AI-generated summaries or drafts before they get used.
- Use of digital signature features is limited to the extent permitted by law.
- Official rules and permissions are applicable to property and land registry services.

06) Deliverables of the Project

- Responsive web application for customers.
- Legal Web App and Mobile Friendly Attorney Assistant Interface.
- Administration portal.
- MySQL database containing sample and test data.
- Authentication, Role Management, Lawyer Verification, Appointment, Document, Client Database, Chatbot and Reporting modules.
- Use Case Diagram, Use Case Descriptions, Component Diagram, ER Diagram .
- Project Proposal document.
- Software Requirements Specification (SRS) document.
- User documentation.
- Technical documentation.
- Test plan, test cases, test report.
- Final project presentation and report

07) Project Constraints and Assumptions

7.1 Constraints

- The project is to be completed by four members of the team within the time-frame of the academic project.

- The main technologies will be HTML, CSS, JavaScript, PHP and MySQL for the first version.
- You will not be guaranteed access to government systems, court databases, official land registry APIs and legal databases.
- Legal document automation should be limited to draft preparation, sharing and support workflows.
- AI and digital signature features are optional / future enhancements due to cost, privacy and legal limitations.
- Legal knowledge content can only be used for the public after being reviewed by lawyers or authorized admins

7.2 Assumptions

- Users access the internet and have suitable devices.
- The lawyers will give the accurate professional information for the check.
- Clients will give correct personal, appointment, and document details.
- Lawyer profiles, document templates and legal knowledge content will be vetted by admins before use by the public.
- Where real legal data is not available, sample data and simulated workflows will be used to test the system.

08) Requirements

8.1 Functional Requirements

This section outlines the functional requirements developers need to implement. The requirements are grouped by module and written using shall statements.

Table 3: Functional Requirements

ID	Requirement	Description	Priority
FR-001	User Registration	Clients, Lawyers, Assistants and Admins will sign up according to their role in the system.	High
FR-002	Login and Authentication	Users need to log in with email/username and password. The system shall restrict functions by user role	High
FR-003	Role-Based Access Control	The system shall allow clients to add and update personal details, contact details and preferences.	High
FR-004	Client Profile Management	The system shall enable lawyers to enter details of their professional profile.	High
FR-005	Lawyer Registration	Admin can verify the details of National Identity Card, Bar and Supreme Court certificate by allowing the lawyers to upload them.	High

FR-006	Lawyer Verification	Admin should have the ability to approve or reject lawyer accounts, or keep them pending.	High
FR-007	Admin Approval	Admins will be able to approve, reject or leave pending lawyer accounts in the system.	High
FR-008	Lawyer Profile	The system shall allow lawyers to keep education, experience, courts, legal categories, consultation fees and availability.	High
FR-009	Legal Category Management	The system should allow the admins to create and update legal categories such as family law, property law, criminal law, civil law, labour law and company law.	High
FR-010	Search and Filter	Clients should be able to search and filter lawyers by location, court, category, experience, availability, rating and fee range.	High
FR-011	Appointment Booking	Clients would be able to schedule lawyer appointments using the system. Lawyers and assistants will be able to accept, reject, reschedule, cancel, and change appointments	High
FR-012	Appointment Management	The system is required to create reminder records and have an option for email/SMS reminders.	High
FR-013	Appointment Reminders	The system will create reminder records and have an option for email/SMS reminders.	Medium
FR-014	Consultation Packages	It should be possible to set up consultation packages based on time, such as 30 minutes, 1 hour, and so forth, by lawyers or administrators.	Medium
FR-015	Payment Record Management	It should log information regarding payment status, receipts, package costs, and service charges..	High
FR-016	Secure Client Database	It should log information regarding payment status, receipts, package costs, and service charges. Every individual lawyer should maintain a private database about each of his/her clients, including contact information, case details, past consultations, and documents.	High
FR-017	Client Data Access Control	A lawyer should not have access to another lawyer's confidential client database unless allowed by the admin/system policy..	High
FR-018	Document Upload and Sharing	It will allow both clients and lawyers/assistants to upload and exchange documents in an allowed case/appointment.	High
FR-019	Document Template Management	The admin/lawyer shall be able to create templates of the documents, and the	Medium

		templates shall only be allowed in the system.	
FR-020	Document Draft Generation	Drafts shall be created by populating the templates with the client's and cases' information.	High
FR-021	Attestation Tracking	The system shall keep track of the documents' status as follows: draft, lawyer reviewed, attested/signed, notarization required, physical signing required, delivery complete, or archived.	High
FR-022	Manual Signing Workflow	The system shall support manual or physical signing workflow for documents not fully executable in electronic form.	High
FR-023	Daily Diary	The system should permit lawyers and assistants to log their daily tasks, court dates, document due dates, client meetings and follow-ups.	High
FR-024	Lawyer Digital Assistant Dashboard	The system shall provide a dashboard showing daily schedule, pending tasks, document review tasks, meeting summaries, and case reminders.	High
FR-025	Meeting Summary Support	The system shall allow lawyers to create meeting summaries manually and shall support optional AI generated summary in future.	Medium
FR-026	Case Management	The system shall allow lawyers to create, update, track and close client matters.	High
FR-027	Case Notes	The system shall support the addition of structured case notes, legal points, next actions and required documents by lawyers.	High
FR-028	Case Outcome Tracking	The system shall allow lawyers to mark case outcomes including won, lost, settled, withdrawn or pending.	Medium
FR-029	Win-Rate Dashboard	The system shall calculate simple statistics from case outcomes stored.	Medium
FR-030	Similar Case Finder	The system shall be able to search stored cases and summaries of case law by category, keywords, legal points, and outcome.	Medium
FR-031	Legal Knowledge Engine	The system shall store approved legal knowledge content such as Constitution basics, fundamental rights, penal law notes, legal notes related to robbery, court hierarchy and procedures.	High
FR-032	Case Law Library	The system shall allow authorized users to save summaries of case law with category, court, year, legal principle and references.	Medium

FR-033	Law Overlap and New Law Notes	The system should support admins/lawyers to add notes on related laws, overlapping legal concepts, law updates for lawyers' reference.	Medium
FR-034	Legal Information Chatbot	The system shall include a chatbot that responds with approved legal knowledge content and a disclaimer that it is general information only.	High
FR-035	Property / Land Registry Request	The system shall enable clients or lawyers to create requests for land-related documents based on district, registry office, deed number/land details, purpose, and service required.	Medium
FR-036	Property Due Diligence Checklist	The system shall produce checklist for property verification like deed copy, survey plan, ownership details, registry extract and missing documents	Medium
FR-037	Land Registry Assistance Workflow	The system shall log the request status i.e. submitted, processing, retrieved, lawyer reviewed, delivered and closed.	Medium
FR-038	Feedback and Rating	The system will also allow clients to rate lawyers and provide feedback after consultations.	Medium
FR-039	Reports and Analytics	The system shall produce reports to users, lawyers, appointments, payments, documents, feedback, cases and service requests.	High
FR-040	Audit Logs	The system will record critical activities such as login, profile update, document upload, approval, payment update and admin updates.	High
FR-041	Admin Content Management	The system will enable the administrators to sustain the legal categories, chatbot content, legal knowledge, document templates and public notices.	High
FR-042	Notification Centre	The system must show notifications for appointments, documents, service requests and admin approvals.	Medium
FR-043	Optional Digital Signature Support	The system shall provide for future integration with LankaSign wherever legally applicable and approved	Low
FR-044	Optional AI Support	The system shall be able to integrate the API of Open Assistant in the future to provide meeting summaries, draft assistance and chatbot enhancement, lawyer review	Low

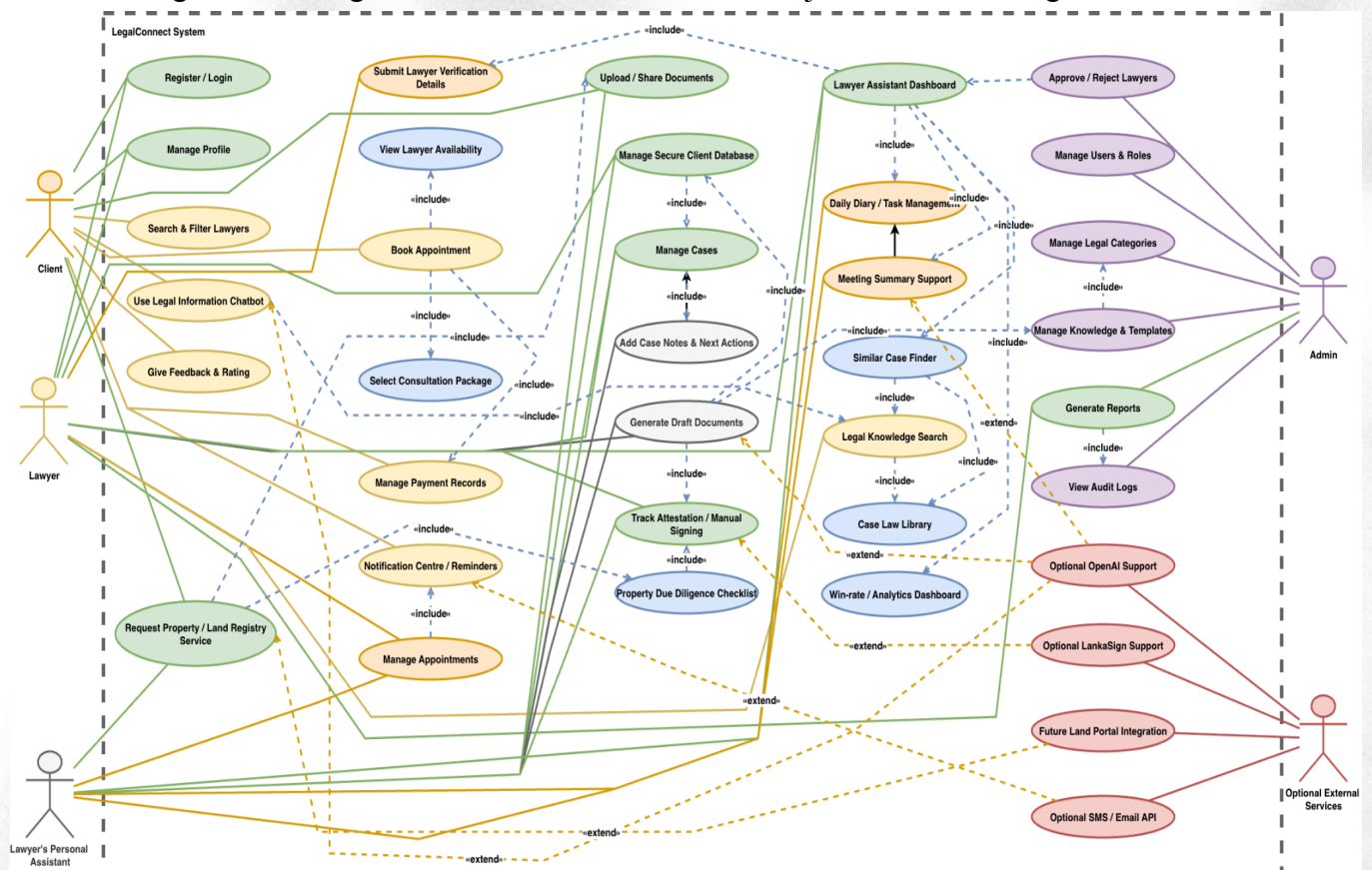
8.2 System Actors

Table 4: System Actors and Descriptions

Actor	Description
Client	Person or organization seeking legal support.
Lawyer	Verified legal professional providing consultation and legal services.
Lawyer's Personal Assistant	Authorized assistant who manages appointments, client records, and documents on behalf of a lawyer.
Admin	Platform manager responsible for verification, content, users, categories, reports, and audit logs.
Optional External Services	Future APIs such as OpenAI, LankaSign, SMS/Email, or official land portal integration where permitted.

8.3 Use Case Diagram

The following use case diagram shows the main actors and major use cases of LegalConnect.



Use Case Diagram available through the following link :

[LegalConnect_Use_Case_Diagram.drawio](#)

8.4 Use Case Description

Table 5: Detailed Use Case Descriptions

Use Case ID	Use Case Name	Main Actor(s)	Description	Include / Extend Relationship
UC-01	Register / Login	Client, Lawyer, Admin	Users can register and log in to the system according to their role. The system validates user credentials and gives access to the correct dashboard.	-
UC-02	Manage Profile	Client, Lawyer	Clients can update personal details. Lawyers can update professional details such as experience, legal categories, courts, fees, and availability.	-
UC-03	Search & Filter Lawyers	Client	Clients can search lawyers based on legal category, location, court area, experience, availability, rating, and fee range.	-
UC-04	Use Legal Information Chatbot	Client	Clients can ask basic legal questions. The chatbot provides general legal information using approved legal knowledge content.	Includes: Legal Knowledge Search. Extended by: Optional OpenAI Support.
UC-05	Request Property / Land Registry Service	Client, Lawyer, Assistant	Users can request land-related support such as deed copy request, extract copy support, or property verification workflow.	Includes: Upload / Share Documents, Property Due Diligence Checklist. Extended by: Future Land Portal Integration.
UC-06	Give Feedback & Rating	Client	Clients can rate lawyers and submit feedback after completing consultations.	-
UC-07	Submit Lawyer Verification Details	Lawyer	Lawyers submit verification information such as NIC, Bar details, and Supreme Court certificate details.	Included by: Approve / Reject Lawyers.
UC-08	View Lawyer Availability	Client	Clients can view available lawyer time slots before booking an appointment.	Included by: Book Appointment.
UC-09	Book Appointment	Client	Clients select a lawyer, available time slot, consultation package, and create an appointment request.	Includes: View Lawyer Availability, Select Consultation Package, Manage Payment Records.
UC-10	Select Consultation Package	Client	Client selects a time-based package such as 30-minute or 1-hour consultation.	Included by: Book Appointment.

Use Case ID	Use Case Name	Main Actor(s)	Description	Include / Extend Relationship
UC-11	Manage Payment Records	Client, Admin	The system records payment status, receipts, consultation package fees, subscription details, or service charges.	Included by: Book Appointment.
UC-12	Notification Centre / Reminders	Client, Lawyer, Assistant	The system shows reminders and notifications for appointments, document updates, property requests, and admin approvals.	Included by: Manage Appointments. Extended by: Optional SMS / Email API.
UC-13	Manage Appointments	Lawyer, Assistant	Lawyers or assistants can accept, reject, reschedule, cancel, and update appointment status.	Includes: Notification Centre / Reminders.
UC-14	Upload / Share Documents	Client, Lawyer, Assistant	Users can upload and share legal documents securely within authorized appointments, cases, or property requests.	Included by: Property / Land Registry Service.
UC-15	Manage Secure Client Database	Lawyer, Assistant	Lawyers can maintain a private client database including client details, case history, documents, notes, and appointment records.	Includes: Manage Cases. Included by: Generate Draft Documents.
UC-16	Manage Cases	Lawyer, Assistant	Lawyers can create, update, track, and close client cases.	Includes: Add Case Notes & Next Actions.
UC-17	Add Case Notes & Next Actions	Lawyer, Assistant	Lawyers can record facts, legal points, required documents, next actions, and follow-up details for a case.	Included by: Manage Cases.
UC-18	Generate Draft Documents	Lawyer, Assistant	The system generates draft legal documents using approved templates and entered client/case data. The document must be reviewed by a lawyer before final use.	Includes: Manage Secure Client Database, Manage Knowledge & Templates, Track Attestation / Manual Signing. Extended by: Optional OpenAI Support.
UC-19	Track Attestation / Manual Signing	Lawyer, Assistant	The system tracks document status such as draft, lawyer reviewed, signed/attested, notary required, physical signing required, delivered, or archived.	Included by: Generate Draft Documents and Property Due Diligence Checklist. Extended by: Optional LankaSign Support.
UC-20	Property Due Diligence Checklist	Lawyer, Assistant	The system supports property verification using a checklist such as deed copy, survey plan,	Includes: Track Attestation / Manual Signing. Included by:

Use Case ID	Use Case Name	Main Actor(s)	Description	Include / Extend Relationship
			ownership details, registry extract, and missing documents.	Property / Land Registry Service.
UC-21	Lawyer Assistant Dashboard	Lawyer, Assistant	The lawyer assistant dashboard provides daily schedule, pending tasks, document review tasks, case reminders, meeting summaries, and analytics.	Includes: Daily Diary, Meeting Summary Support, Similar Case Finder, Win-rate / Analytics Dashboard.
UC-22	Daily Diary / Task Management	Lawyer, Assistant	Lawyers or assistants can manage court dates, client meetings, document deadlines, reminders, and follow-up tasks.	Included by: Lawyer Assistant Dashboard.
UC-23	Meeting Summary Support	Lawyer, Assistant	Lawyers can create structured meeting summaries including client issue, key facts, required documents, next steps, and follow-up dates.	Included by: Lawyer Assistant Dashboard. Extended by: Optional OpenAI Support.
UC-24	Similar Case Finder	Lawyer	The system helps lawyers search previous similar cases, case notes, and case law summaries based on legal category, keywords, and legal points.	Includes: Legal Knowledge Search, Case Law Library.
UC-25	Legal Knowledge Search	Client, Lawyer	Users can search approved legal information such as legal categories, court hierarchy, fundamental rights, penal law notes, and general legal procedures.	Includes: Case Law Library. Included by: Chatbot and Similar Case Finder.
UC-26	Case Law Library	Lawyer, Admin	Authorized users can store and search case law summaries with case name, court, year, legal principle, category, and references.	Included by: Similar Case Finder and Legal Knowledge Search.
UC-27	Win-rate / Analytics Dashboard	Lawyer	The system displays simple case outcome statistics such as won, lost, settled, pending, and category-wise case counts based on stored data.	Included by: Lawyer Assistant Dashboard.
UC-28	Approve / Reject Lawyers	Admin	Admin reviews lawyer verification details and approves, rejects, or keeps the lawyer account pending. Approved lawyers become visible to clients.	Includes: Submit Lawyer Verification Details.
UC-29	Manage Users & Roles	Admin	Admin can manage clients, lawyers, assistants, and system roles with access control.	-
UC-30	Manage Legal Categories	Admin	Admin can add, update, or remove legal categories such as family law, property law, criminal law, civil law, labour law, and company law.	Included by: Manage Knowledge & Templates.

Use Case ID	Use Case Name	Main Actor(s)	Description	Include / Extend Relationship
UC-31	Manage Knowledge & Templates	Admin, Lawyer	Admin or authorized lawyers can manage legal knowledge content and approved document templates used by chatbot and document generation.	Includes: Manage Legal Categories. Included by: Generate Draft Documents.
UC-32	Generate Reports	Admin, Lawyer	The system generates reports for appointments, payments, feedback, documents, cases, property requests, lawyer performance, and system usage.	Includes: View Audit Logs.
UC-33	View Audit Logs	Admin	Admin can view important system activities such as login, profile updates, document uploads, approvals, payments, and admin changes.	Included by: Generate Reports.
UC-34	Optional OpenAI Support	Optional External Services	Future optional API support for chatbot improvement, meeting summary generation, and draft document assistance.	Extends: Chatbot, Meeting Summary Support, Generate Draft Documents.
UC-35	Optional LankaSign Support	Optional External Services	Future optional digital signature support for suitable documents where legally permitted.	Extends: Track Attestation / Manual Signing.
UC-36	Future Land Portal Integration	Optional External Services	Future integration with official land registry/land portal services if legal and technical permission is available.	Extends: Property / Land Registry Service.
UC-27	Optional SMS / Email API	Optional External Services	Future optional notification service for sending appointment reminders and status alerts through SMS or email.	Extends: Notification Centre / Reminders.

8.5 Database Design Overview

The following tables are proposed for the first database design. The exact fields can be refined during detailed database design.

Table 6: Database Schema Overview

Table	Purpose
users	Stores common login details and user role.
clients	Stores client profile details.
lawyers	Stores lawyer profile and professional details.
lawyer_verification	Stores NIC, Bar details, Supreme Court certificate details, and approval status.
assistants	Stores lawyer assistant details and assigned lawyer.
legal_categories	Stores legal service categories and subcategories.
lawyer_categories	Maps lawyers to legal categories.
availability_slots	Stores lawyer available dates and times.

appointments	Stores bookings, status, time, package, and related client/lawyer.
consultation packages	Stores time-based consultation packages and fees.
payments	Stores payment records, receipts, and status.
client records	Stores personalized client database entries per lawyer.
cases	Stores case details, status, outcome, and related client/lawyer.
case notes	Stores facts, legal points, notes, and next actions.
documents	Stores document metadata and file paths.
document templates	Stores approved document templates.
document attestation	Stores review, signature, attestation, and physical signing status.
daily diary	Stores lawyer tasks, court dates, follow-ups, and reminders.
meeting summaries	Stores manual or AI-assisted meeting summaries.
knowledge_base	Stores legal information content approved for chatbot and research assistance.
case laws	Stores case law summaries and legal principles.
law overlap notes	Stores related law references, updates, and overlap notes.
property requests	Stores land registry/property document request details and status.
property checklists	Stores due diligence checklist results.
feedback	Stores client ratings and feedback.
notifications	Stores system notifications.
audit_logs	Stores important system actions for monitoring and accountability.

8.6 Quality Attributes Requirements

Table 7: Quality Attribute Implementation

Quality Attribute	How it will be achieved
Security	Passwords will be hashed, role-based access will be used, and sensitive actions will be logged.
Privacy	Client records and legal documents will be accessible only to authorized users.
Confidentiality	Case notes, documents, and meeting summaries will not be publicly visible.
Usability	Interfaces will use clear navigation and simple language suitable for users with basic technical knowledge.
Performance	Common pages should load within acceptable time under normal testing conditions.
Maintainability	The system will be divided into modules such as users, appointments, documents, knowledge base, reports, and admin.
Scalability	The database design will support future growth in users, documents, appointments, and legal content.
Reliability	The system will validate input and handle errors without losing user data.
Legal Safety	Chatbot and document outputs will clearly state that they are general information or draft content and require lawyer review.

09) Technologies to be Used

Table 8: Technology Stack

Category	Technology	Purpose
Frontend	HTML, CSS, JavaScript, Bootstrap	Web pages, styling, responsive interface, validation, and dynamic behaviour.
Backend	PHP	Server-side processing, authentication, business logic, forms, file upload, and reports.
Database	MySQL	Store users, lawyers, clients, appointments, cases, documents, payments, knowledge content, and reports.
Server / Local Environment	Apache / XAMPP	Local development and testing.
Design and Diagrams	Draw.io / diagrams.net	Use case, component, ER, and activity diagrams.
Collaboration	GitHub, Google Docs	Version control, documentation, and team collaboration.
Testing	Manual test cases, browser developer tools	Functional testing, validation, and defect tracking.

9.1 API or Library Usage with Justification

Table 9: API and Library Justifications

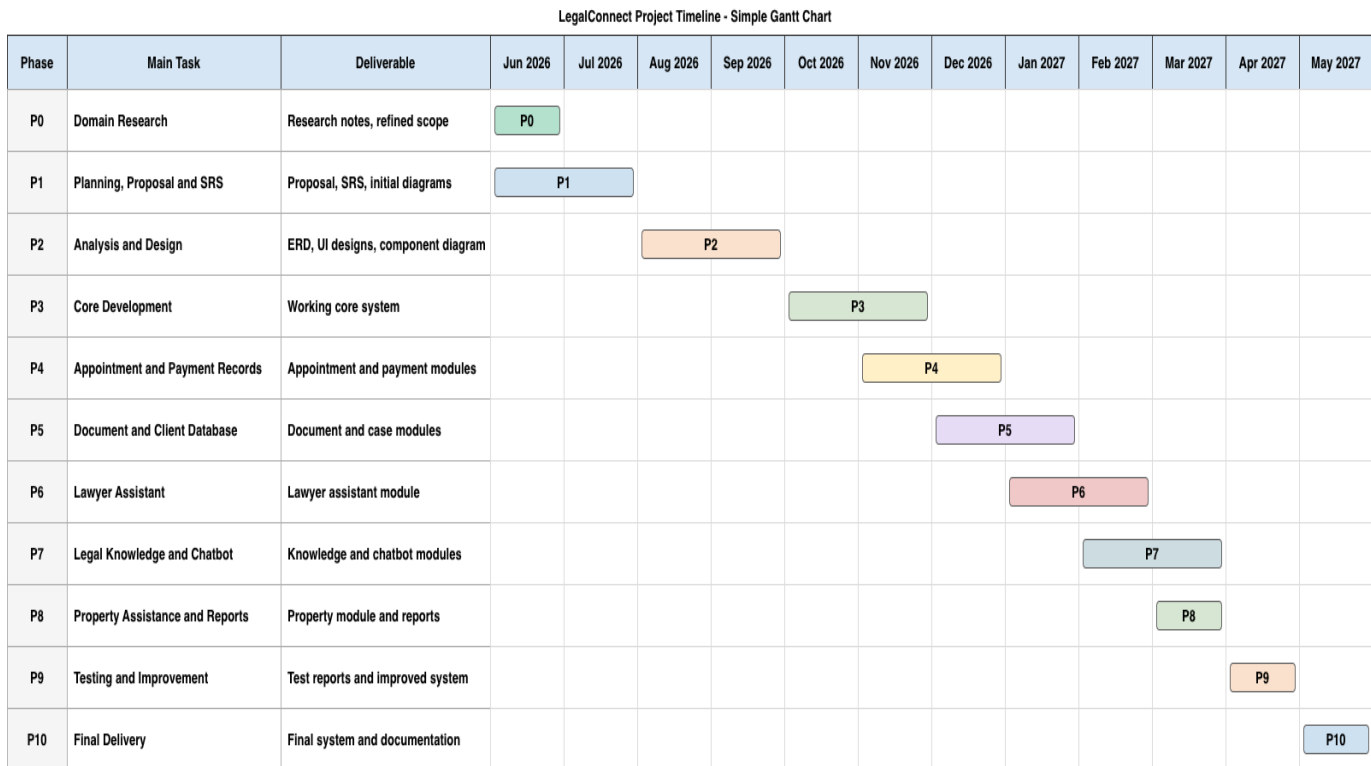
API / Library	Usage	Justification
No mandatory external API	Core first version	Keeps the first version feasible and reduces dependency risk.
PDF generation library such as TCPDF/DomPDF	Reports, receipts, draft documents	Useful for generating legal and administrative documents.
OpenAI API	Future chatbot enhancement, meeting summaries, draft assistance	Optional innovation; outputs must be reviewed by lawyers because legal data is sensitive.
LankaSign	Future digital signature support where legally valid	Useful only for documents where digital signing is legally suitable.
SMS/Email API	Future appointment reminders and status alerts	Improves real-world usability.

10) Project Timeline

The project will follow an Agile incremental development methodology. The team will develop the system in phases, starting with core modules and gradually adding advanced features. The timeline matches the selected incremental development model.

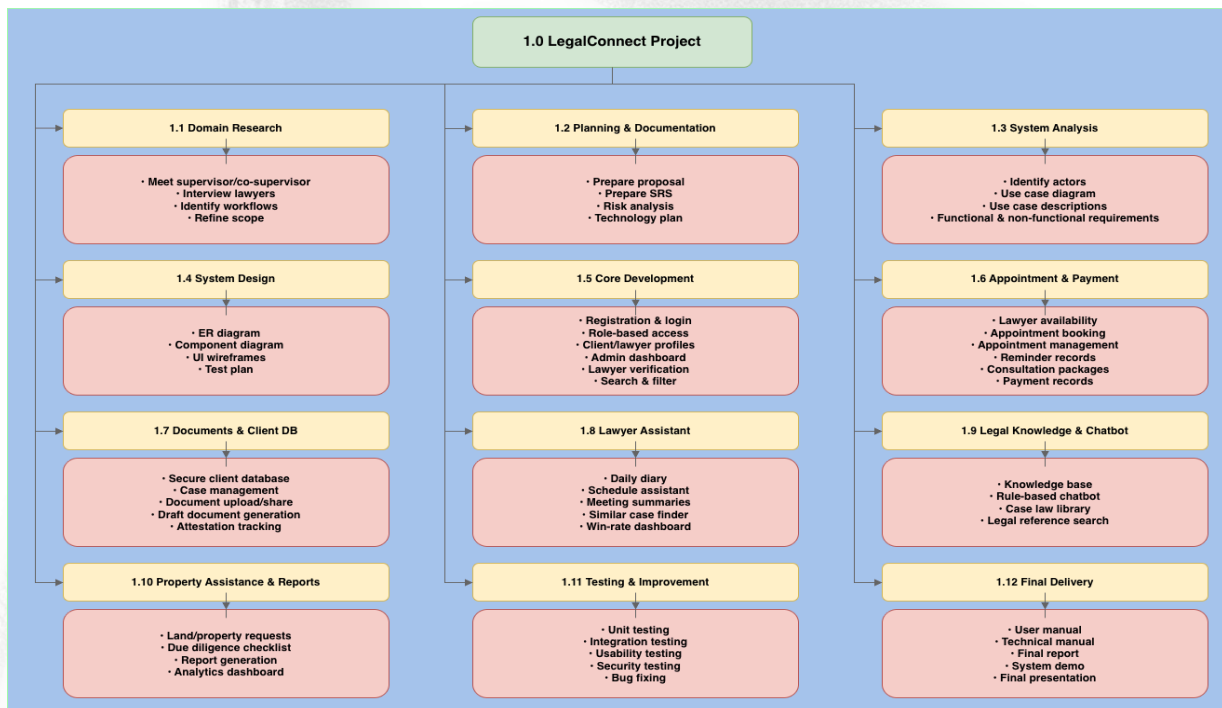
Table 10: Project Phase Timeline

Phase	Time Period	Main Activities	Deliverables
Phase 0: Domain Research	Month 1	Meet supervisor and lawyers; validate legal workflows, document handling, attestation, and property/land registry process.	Research notes and refined scope
Phase 1: Planning and Proposal/SRS	Month 1-2	Prepare proposal, SRS, use case diagram, requirements, risk analysis, and technology plan.	Proposal, SRS, initial diagrams
Phase 2: Analysis and Design	Month 3-4	Design database, UI wireframes, system architecture, module responsibilities, and test plan.	ERD, UI designs, component diagram
Phase 3: Core Development	Month 5-6	Develop login, roles, client profile, lawyer profile, admin portal, verification, categories, and search/filter.	Working core system
Phase 4: Appointment and Payment Records	Month 6-7	Develop availability, appointment booking, reminders, consultation packages, and payment record management.	Appointment and payment modules
Phase 5: Document and Client Database	Month 7-8	Develop client database, case management, document upload, template draft generation, and attestation tracking.	Document and case modules
Phase 6: Lawyer Assistant	Month 8-9	Develop daily diary, schedule assistant, meeting summaries, similar case search, and win-rate dashboard.	Lawyer assistant module
Phase 7: Legal Knowledge and Chatbot	Month 9-10	Develop knowledge base, rule-based chatbot, case law library, and legal reference search.	Knowledge and chatbot modules
Phase 8: Property Assistance and Reports	Month 10	Develop land/property request workflow, due diligence checklist, reports, and analytics.	Property module and reports
Phase 9: Testing and Improvement	Month 11	Conduct unit, integration, usability, security, and acceptance testing; fix bugs.	Test reports and improved system
Phase 10: Final Delivery	Month 12	Prepare final documentation, user manual, technical manual, final report, demo, and presentation.	Final system and documentation



Gantt Chart available through the following link: [LegalConnect_Gantt_Chart.drawio](#)

10.1 Work Breakdown structure



Wbs available through the following link: [LegalConnect_WBS_Diagram.drawio](#)

11) Declaration

We, as members of the project titled LegalConnect: A Smart Legal Consultation and Lawyer Practice Management Platform for Sri Lanka, certify that we will carry out this project according to the guidelines provided by the coordinators and supervisors of the course. We also certify that we will not incorporate, without acknowledgement, any material previously submitted for a degree or diploma in any university. To the best of our knowledge and belief, the project work will not contain any material previously published or written by another person or ourselves except where due reference is made in the appropriate places.

Index Number	Name of the Student	Signature
24020206	J.B.D Silva	
24021113	P.D.V.R Weerakoon	
24020818	O.S Premachandra	
24020664	P. M. P. Methsarani	

Table 11: Member Declarations and Signatures

12.) References

- [1] SCS2202/IS2102 - Group Project I 2026 Project Proposal Guidelines.
- [2] Project discussion notes from supervisor/co-supervisor meetings and lawyer requirement gathering discussions.